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Sensitive Surface Enhanced Raman Scattering-Based Detection of a *BIGH3* Point Mutation Associated with Avellino Corneal Dystrophy

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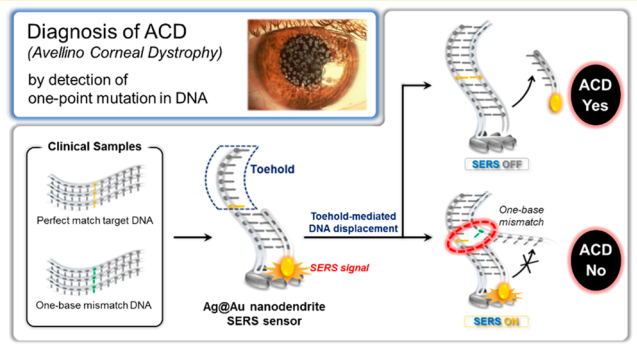
Supporting Information

ABSTRACT: Surface enhanced Raman scattering (SERS) is highly useful for sensitive analytical sensing; however, its practical availability for detecting a point mutation associated with disease in clinical sample was rarely proved. Herein, we present a toehold-mediated, DNA displacement-based, SERS sensor for detecting point mutations in the *BIGH3* gene associated with the most common corneal dystrophies (CDs) in a clinical setting. To diagnose Avellino corneal dystrophy (ACD), selectivity was ensured by exploring optimal DNA displacement conditions such as length of toehold and hybridization temperature. A SERS-efficient Ag@Au bimetallic nanodendrite was employed to ensure sensitivity. Optimization for a clinical setting showed that discrimination was maximized when toehold length was 6-mer (T6), and hybridization temperature was 36 °C. On the basis of tests that used clinical homozygous and heterozygous CD samples, a single-base mismatched DNA sequence was identifiable within 30 min with a limit of detection (LOD) of 400 fM. From the results, we conclude that our toehold-mediated, DNA displacement-based, SERS sensor allows a rapid and sensitive detection of a *BIGH3* gene point mutation associated with Avellino corneal dystrophy, indicating the practical ability of the method to diagnose genetic diseases caused by point mutations.



Point mutations in DNA that change the amino acid sequence of proteins can cause serious genetic disorders and cancers.[1−3] Most common corneal dystrophies (CDs) such as Avellino corneal dystrophy (ACD), Reisec-Bucklers CD (RBCD), and lattice CD type I (LCD) are caused by amino acid substitutions resulting from point mutations in the *BIGH3* gene (Table 1).[4,5] Case reports indicate that laser-assisted in situ keratomileusis (LASIK) aggravates corneal deposits and causes repeated recurrence of symptoms in ACD patients.[6−10] Therefore, especially for patients with a heterozygous ACD genotype, ACD must be diagnosed before LASIK to prevent blindness; they usually do not show noticeable eyes symptoms before they are in their 60s.[7,9]

Table 1. Single Point Mutations in BIGH3 Associated with Corneal Dystrophies

DNA sequence	protein	phenotype
CGC to CAC (418)	R124H	Avellino corneal dystrophy
CGC to CTC (418)	R124L	Reisec-Bucklers corneal dystrophy
CGC to TGC (417)	R124C	Lattice corneal dystrophy

A simple sensor that could identify single-point mutations in DNA in different genotypes would be beneficial for practical clinical analysis since standard methods including DNA sequencing are time-consuming and labor intensive. Advances in the sensing technique that uses diverse signal tagging, such as fluorescence,[11,12] electrochemistry,[13] surface plasmon resonance (SPR),[14,15] colorimetry,[16,17] and surface-enhanced Raman scattering (SERS),[18,19] have been reported. However, results have been mostly confined to detection of model DNAs. Substantial improvements in analytical performance such as sensitivity are required for clinical applications.

Here, we report a SERS sensor that incorporated a toehold-mediated DNA displacement reaction to identify a single-base mismatch (1MM) in hot-spot mutation regions of the *BIGH3* gene for diagnosis of ACD. A Ag@Au bimetallic nanodendrite (Figure S1)[20] was used as a SERS platform to induce a large signal change upon binding of a target DNA (tDNA) (Figure 1).

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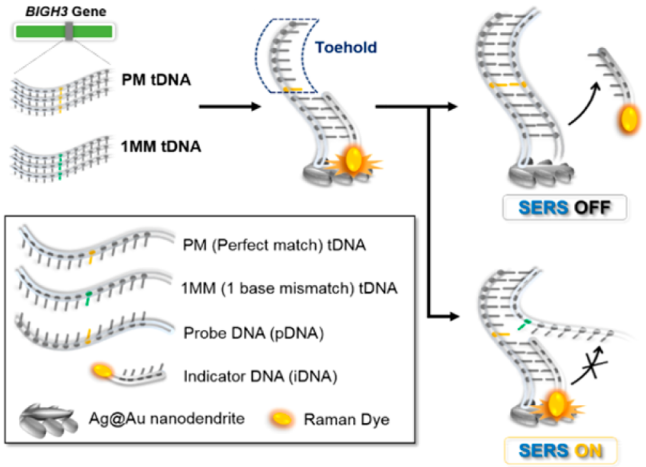

Figure 1. Overall scheme for SERS-based detection of perfect match (PM) tDNA against single-base mismatch (1MM) tDNA to pDNA using an Ag@Au nanodendrite.

A toehold-mediated DNA-displacement reaction, employing an unhybridized base region (the toehold) at the end of a double-stranded duplex to trigger DNA hybridization, is advantageous for DNA detection, since it provides a thermodynamic driving force for hybridization and so allows the reaction to proceed at room temperature without enzymatic assistance.[20−23] As the sensor, a double-stranded DNA (dsDNA) composed of a probe DNA (pDNA) and an indicator DNA (iDNA) with a Raman reporter (Cy5) is immobilized on the Ag@Au nanodendrite. The pDNA sequence is a perfect match (PM) to a tDNA, and the iDNA does not cover the location of the point mutation to provide selectivity for recognition of the tDNA. Conjugation of the duplex on the nanodendrite locates the Raman reporter near the surface and activates the Raman signal. Access of the tDNA to the pDNA forces the prehybridized iDNA away from the strand through toehold-mediated displacement. The release of iDNA decreases the Raman signal. In the presence of 1MM tDNA, iDNA is not released and the initial Raman signal is maintained.

Our SERS-based sensor is a binary sensing system that identifies target information by using four pDNAs, normal, ACD, RBCD, and LCD, which detects whether PM tDNA to a pDNA is present in a given sample. One of most important analytical merits of the sensor is its ability to differentiate PM tDNA from 1MM tDNA based on pDNAs. The *BIGH3* point mutation (ACD, RBCD, or LCD tDNA) will show a decreased signal on the respective pDNA (ACD, RBCD, or LCD pDNA) but not on normal pDNA in which 1MM tDNA is applied to normal pDNA.

Since the kinetics of strand displacement depends on the length and sequence of the toehold,[22,23] 2-, 4-, 6-, and 8-mer toehold structures (designated T2, T4, T6, and T8, respectively) were tested. To make these toeholds, 14-, 16-, 18-, and 20-mer pDNAs were prepared and a 11-mer iDNA was prehybridized on the second through the 12th bases on the pDNA (Figure 2A). Sequences of pDNAs, iDNA, and tDNA are in Table S1. Percent intensity changes (PICs) before and after measurement of a 20 nM PM tDNA sample (20-mer) were calculated using the 1468 cm^{-1} Cy5 band. PICs measured for a 1MM tDNA sample and buffer (no tDNA) were also compared. For T2, discrimination between the PM tDNA and

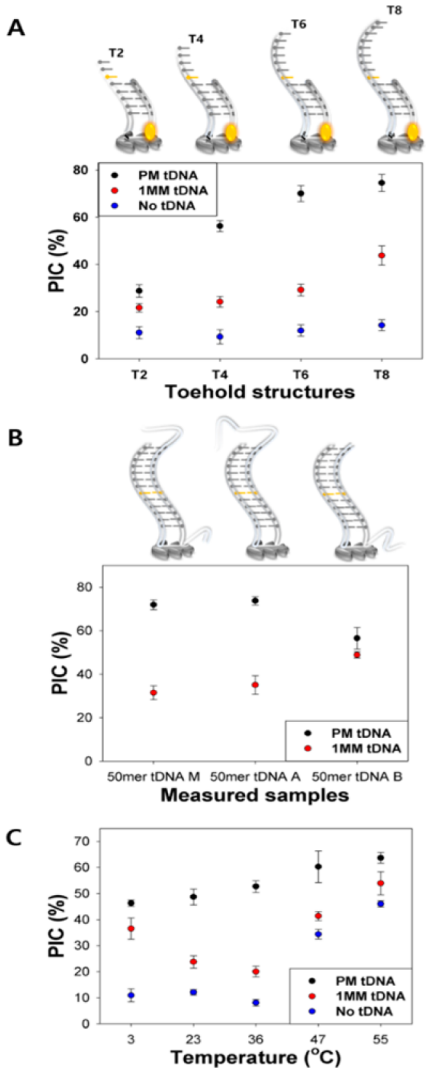

Figure 2. Optimization of point mutation-sensing using synthesized tDNA. PICs (black circles) occurred when toehold length was 2-mer to 8-mer for measurement of a PM tDNA sample (A), for measurement of three synthetic 50-mer tDNAs with recognition sites at different locations (B), under hybridization temperatures at 4, 23, 36, 47 and 55 °C for measuring 20-mer PM tDNA samples (C). PICs for measuring a 1MM tDNA sample (red circles) and buffer containing no tDNA (blue circles) were compared. In (A, B), 1 h of hybridization was performed at room temperature with 20 nM tDNA. In (C), hybridization was performed for 30 min with 200 pM tDNA.

1MM tDNA was unsatisfactory because of the short toehold length. As toehold length increased, PICs increased for measuring PM tDNA. This result clearly indicated that the longer toehold provided greater driving force to attract the tDNA for subsequent release of the iDNA. Discrimination was maximized with the 18-mer pDNA and 11-mer iDNA pair (T6).

With real DNA samples, tDNAs longer than 50-mers are typically used. The position of the toehold recognition site in a long tDNA could affect detection performance. Three synthetic 50-mer tDNAs with recognition sites at different locations were prepared using the T6 structure (Figure 2B). The 50-mer M had the recognition site approximately in the middle, so the upper and lower parts became free DNA chains after hybridization. The 50-mer A and 50-mer B had a recognition site in the lower and upper part of the tDNA, respectively. In

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measurements using PM 50-mer M and PM 50-mer A, discrimination of the corresponding 1MM tDNAs was clear. PIC for measurement of 1MM 50-mer B was similar to that for measurement of PM 50-mer B. The longer free chain at the bottom of 1MM 50-mer B more easily adsorbed to the nanodendrite surface as reported,[24] providing a strong thermodynamic driving force to displace the iDNA, thereby ignoring the presence of 1MM sequence. Therefore, a tDNA with the toehold-recognition site in the middle could result in a balanced thermodynamic state made by free DNA chains at both sides, after recognition by and hybridization to the toehold region.[25]

The optimal temperature to induce maximal detection sensitivity was determined. Using the 20-mer tDNA and T6 toehold structure, PICs at hybridization temperatures of 4, 23, 36, 47, and 55 °C were examined (Figure 2C). At 23 °C, 1MM discrimination was feasible; however, hybridization at 36 °C maximized 1MM discrimination. At temperatures higher than 36 °C, PICs for measurements of PM tDNA increased, while the 1MM tDNA also induced a large signal change due to the lower melting temperature (T_m) of the iDNA. The calculated T_m of the iDNA was 57 °C (Figure S2), explaining the nonspecific release of iDNA at elevated temperatures. The PIC value of 1MM tDNA at low temperature (3 °C) increased possibly because thermal energy was too low to discriminate the 1MM sequence. In the hybridization process, there are several partial stable states and partial hybridization states.[26] Thermal energy may be too low to overcome the energy barrier of the partial hybridization state at 3 °C. Therefore, DNA hybridization potentially occurs under this condition, preventing discrimination of the 1MM sequences and causing the PIC of 1MM tDNA at 3 °C to increase. A reaction performed over 30 min (10, 20, 30, 45, and 60 min were tested) at 36 °C maximized 1 MM discrimination. All subsequent DNA hybridizations were executed at 36 °C over 30 min.

Five purified and unpurified real samples containing tDNAs with varying chain lengths of 50-, 100-, 147-, 190-, and 225-mer were prepared (Table S2) and measured at optimal conditions. The concentration of tDNA was 10 nM, corresponding to a 10^5-fold dilution of the PCR-amplified sample (Figure 3A). Discrimination between PM (circles) and 1MM tDNAs (triangles) improved as tDNA length increased up to 147-mer. Discrimination degraded at longer lengths because the intermolecular secondary structures of tDNA might have influenced DNA hybridization, as previously reported.[15] The 147-mer tDNA gave the best discrimination for measuring both purified and unpurified samples. The measurement was reliable and not obstructed by enzymes, salt, or primers in the unpurified samples. The limit of detection (LOD) calculated from the response curve of tDNA concentrations (100 fM to 100 nM range) vs PICs was approximately 400 fM (Figure 3B). The superior reproducibility (average relative standard deviation (RSD): 10.2%) and sensitivity were attributed to the use of the Ag@Au bimetallic nanodendrite, which incorporated the advantages of Au for stability and Ag for sensitivity in the dendrite structure. Also, it provided many hot spots because of high population of edges and angles in terraces.[27,28]

BIGH3-associated CD is an autosomal dominant disorder that requires only a single mutated allele to cause the disease. In clinical situations, genetic mutations in either allele must be identified. Therefore, normal and homozygous or heterozygous ACD, RBCD, and LCD samples were used to assess diagnostic

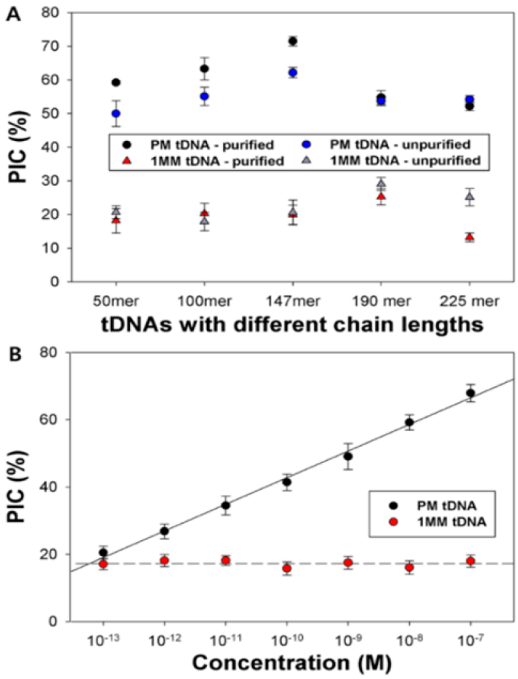

Figure 3. (A) PICs results from measurements of purified and unpurified real DNA samles (50-, 100-, 147-, 190-, and 225-mer). PM (circles) and 1MM (triangles) tDNAs were measured. (B) Variation of PICs in measuring 1 fM to 100 nM PM tDNA samples. The results from corresponding 1 MM tDNA samples with the same concentration range were compared. Hybridization was performed with 10 nM tDNA for 30 min at 36 °C.

performance. Homozygous meant a genotype with two identical alleles, and heterozygous meant two different alleles at a given locus. It is important to diagnose heterozygous samples because inevitable visual loss can occur after LASIK in heterozygous ACD patients with ambiguous symptoms. It may be also important to distinguish the presence of 1MM tDNA and No tDNA because iDNA is not released and the Raman signal is maintained in both cases. This may be important when tDNA is not amplified, which can lead to misdiagnosis as 1MM tDNA. Our sensor avoids this issue by testing four probes simultaneously. In the case of No tDNA, no signal changes were observed for any of the probes, whereas there will be a signal change on at least one of the probes in the case of 1MM tDNA.

Real tDNA samples prepared from study participant blood samples were measured. To examine participants' DNA using this system, we amplified the 147-mer region containing the mutated site from the participants' genomic DNA as described in the Supporting Information. PICs over 30% were classified as Yes (Y), and others were classified as No (N). Amplified tDNA samples from homozygous patients will show a signal only for the respective probe for the corresponding corneal disease. However, tDNA samples amplified from heterozygous patients will show a signal for both the normal and respective probe for the disease because heterozygous samples contain one normal allele and one mutated allele. Results of hybridization stringency tests using each pDNA type to measure four homozygous tDNAs (normal, ACD, RBCD, LCD) are shown in Figure 4A. Signals were Y only when the pair was matched, such as for normal pDNA−normal tDNA, ACD pDNA−ACD tDNA, RBCD pDNA−RBCD tDNA, and LCD pDNA−LCD tDNA. This result confirmed that the method discriminated 1

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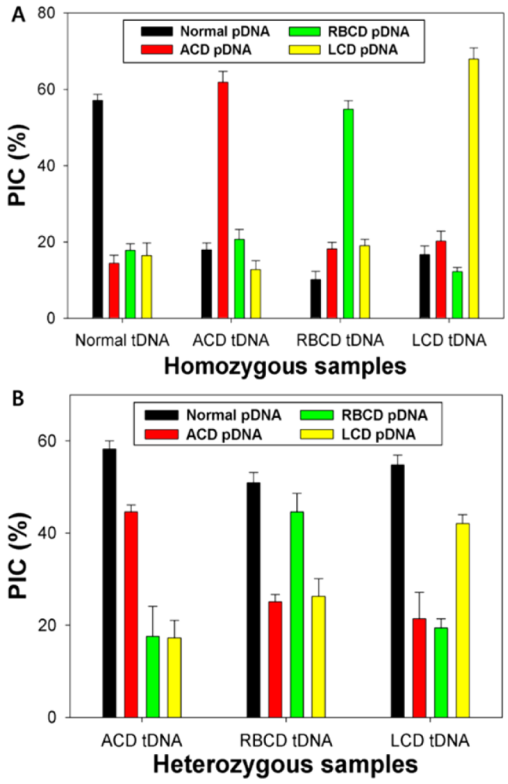

Figure 4. Hybridization stringency tests using normal, ACD, RBCD, and LCD pDNA for measuring amplified real homozygous (A) and heterozygous (B) tDNA samples. Hybridization was performed with 10 nM 147-mer tDNA of homozygous or heterozygous sample for 30 min at 36 °C.

MM in the *BIGH3* gene. In real tDNA samples, finding 1MM in heterozygous patients is important since patient samples are mostly heterozygous. PICs were expected to be Y for a normal pDNA and a corresponding pDNA for the measurement. The signal of heterozygous CD samples was Y for both the normal and corresponding CD pDNAs (Figure 4B). This result indicated the diagnostic ability for heterozygous and homozygous CDs using clinical samples.

In conclusion, we have presented a type of Ag@Au bimetallic SERS sensor for rapid and sensitive detection of DNA point mutations in the *BIGH3* gene causing common CDs. This bimetallic SERS sensor could successfully detect homozygous and heterozygous mutations causing ACD, RBCD, and LCD with excellent reproducibility (RSD = 10.2%) and with negligible interference of solution matrix (e.g., enzyme, proteins, primers, salt, etc.), high sensitivity (ca. 400fM), and a rapid rate (<30 min). This is, as we believe, the first example of detecting DNA point mutation and diagnosing genetic disease using the method, which is superior to the previously reported point-mutation detection using a SERS system with real samples. Taken together, a well-defined Ag@Au dendrite structure and careful optimization of the DNA displacement-based detection of perfect-matched DNA hybridization in this study was successfully accomplished for successful identification of each CD. Therefore, our SERS sensor would be useful as a new diagnostic platform that allows selective and sensitive detection of various DNA point mutations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.6b03320.

Experimental section; table of probe, indicator, and target DNA oligonucleotides; table of primers; SEM image of Ag−Au nanodendrites; experimental melting curve of iDNA (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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